

Yuvraj Sehgal

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AI/ML Engineer and M.Sc. Computer Science researcher with 4+ years of software engineering experience spanning production AI systems, deep learning, LLM/RAG applications, data-intensive backends, and AI for systems. Author of peer-reviewed research published at ACM FSE and the *Journal of Systems and Software*. Experienced in building end-to-end ML and AI solutions using Python, PyTorch, NLP, LLMs, LangChain/LangGraph, knowledge graphs, and cloud infrastructure.

Technical Skills

Languages: Python, SQL, Java, JavaScript/TypeScript, Bash

AI & Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, deep learning, NLP, LLMs, LLM fine-tuning, prompt engineering, RAG, LangChain, LangGraph

ML & AI Systems: Agentic AI, ReAct, knowledge graphs, model inference, AI-assisted root-cause analysis, anomaly detection, synthetic data generation, diffusion models

Data Engineering: Apache Spark, Databricks, Apache Airflow, dbt, Pandas, NumPy, ETL/ELT, data pipelines, dimensional modeling, data warehousing, MapReduce

Databases: PostgreSQL, AlloyDB, MongoDB, Firestore, Neo4j

Cloud & Infrastructure: GCP (Cloud Run, Compute Engine), Docker, Kubernetes, Terraform, CI/CD, Jenkins, Git, Bitbucket

Backend & APIs: FastAPI, Node.js, Spring Boot, REST APIs

Testing & Observability: JUnit, Jest, Selenium, Appium, RestAssured, Cucumber/BDD, React Testing Library, OpenTelemetry

Education

M.Sc. Computer Science | Brock University

Sept 2025 – Sept 2027 (expected)

Thesis-based; research in AI for systems and trustworthy, responsible AI | St. Catharines, ON

B.Sc. (Honours) Computer Science, Intelligent Systems | Brock University

Jan 2021 – Jan 2025

GPA: 87/100 | Dean's Honour List (all years) | Golden Key International Honour Society

Experience

Artificial Intelligence Researcher | Brock University

Feb 2026 – Present

Mitacs Accelerate industrial fellowship with Ciena Corp. | St. Catharines, ON

- **TraceSynth:** Developed constraint-guided diffusion models for generating production-quality Linux kernel traces across multiple trace channels; published at *ACM FSE 2026 (Industry Track)*.
- Developed and evaluated Transformer-based denoising diffusion models for synthetic system-trace generation, achieving **87.2% F1-Macro** on downstream system-diagnostic classification and improving performance by **104%** at 4,096-token sequence length compared with shorter-context generation.
- Applied constraint-guided repair to enforce structural and system-level trace validity, improving downstream performance by up to **4.3 percentage points**; investigated reduced-channel models that retained **97–99%** of full-channel performance at substantially lower computational cost.
- **LMAT:** Contributed to an adaptive tracing framework using language models for efficient system-behavior analysis; published in the *Journal of Systems and Software (2026)*.
- Built an agentic incident-triage system using **LangGraph**, **ReAct**, **LLM-based reasoning**, **telemetry**, **logs**, and **historical Jira data** to support automated root-cause diagnosis and incident retrieval.
- Curated and processed large-scale microservice trace and log datasets from **DeathStarBench** and **TrainTicket** on GCP to support AIOps, incident analysis, and machine-learning research.

AI/ML Engineer | Mengalo

Aug 2024 – Dec 2025

Toronto, ON

- Designed, developed, and deployed production **AI services and REST APIs** using Python and FastAPI, containerized with Docker and deployed on GCP for business and customer-engagement automation.
- Built an **LLM- and RAG-powered analytics platform** using LangChain and Neo4j, combining natural-language querying, retrieval, and knowledge-graph representations to interact with operational data.
- Developed NLP and LLM pipelines for **prompt engineering**, **information extraction**, **semantic processing**, and **knowledge-graph construction**, integrating AI capabilities into production backend services.
- Designed data-processing workflows connecting application data, PostgreSQL, and Neo4j to support reliable retrieval and AI-driven analytics while maintaining controlled database access patterns.

Research Assistant | Brock University

Jan 2024 – Jan 2025

St. Catharines, ON

- Developed **AURA (AI-powered Unified Research Artifact Evaluation)**, an AI-driven pipeline for automated evaluation and analysis of research artifacts using **Python, LLMs, OpenAI APIs, LangGraph, Neo4j, and vector search**.
- Designed an end-to-end **RAG and knowledge-graph architecture** combining semantic retrieval, structured artifact meta-data, and LLM-based reasoning to automate evidence extraction and artifact evaluation across **2,000+ research artifacts from 40 conferences**.
- Built automated data-ingestion and document-processing pipelines to extract and normalize **30,000+ Markdown documents**, transforming unstructured research artifacts into searchable representations for downstream AI analysis.
- Developed reusable evaluation workflows for extracting artifact capabilities, requirements, and supporting evidence, enabling systematic analysis of research reproducibility and artifact quality.

Data Engineer | Scotiabank

May – Aug 2024

Toronto, ON

- Built an end-to-end **Databricks-to-AlloyDB migration pipeline** for up to **350 billion rows**, covering data extraction, transformation, validation, loading, reconciliation, and performance optimization; improved downstream query performance by approximately **10x**.
- Designed and implemented an **incremental data-processing pipeline** that reduced daily Databricks processing time from approximately **6 hours to 2 hours**, cutting compute usage and accelerating downstream data availability.
- Developed and maintained production **Apache Airflow DAGs in Astronomer** for scheduled and incremental data workflows, integrating Databricks jobs, GCP Cloud Storage buckets, and downstream processing stages.
- Worked with **Databricks Unity Catalog** for governed data access and pipeline management, and supported production deployments through **CI/CD** workflows while maintaining scheduled Databricks jobs and data pipelines.

Full Stack Developer | Scotiabank

May – Aug 2023

Toronto, ON

- Built and deployed the bank's LMS platform using React, Node.js, REST-based backend microservices, Firestore, and GCP Cloud Run.
- Developed automated JUnit, Jest, and React Testing Library coverage, increasing test coverage by **15 percentage points** to meet the bank's 85% quality mandate.

QA Automation Engineer | Scotiabank

Sept – Dec 2022

Toronto, ON

- Automated 35+ regression scenarios for front-office financial systems using **Java, Selenium, and Cucumber BDD**, reducing manual testing effort by approximately **70%**.
- Integrated automated test suites into **Jenkins CI/CD** workflows using Git and Bitbucket to support continuous validation and code review.
- Built automated Android and API test suites using **Java, Selenium, Appium, and RestAssured**, covering 15+ scenarios and reducing testing time by over **75%**.
- Developed reusable API automation and reporting components, reducing validation time by approximately **50%**.

Publications

Y. Sehgal, S. Patel, M. Panahandeh, N. Ezzati-Jivan, F. Tetreault. **TraceSynth: Generating Production-Quality Kernel Traces with Constraint-Guided Diffusion Models**. *ACM FSE Companion '26, Industry Track*, ACM, pp. 496–506. DOI: 10.1145/3803437.3805222

K. Darvishi, M. Noferesti, Y. Sehgal, N. Ezzati-Jivan. **LMAT: An Adaptive Tracing Approach Based on Efficient System Behavior Analysis Using Language Models**. *Journal of Systems and Software*, vol. 238, art. 112890, Elsevier, 2026. DOI: 10.1016/j.jss.2026.112890

Awards & Leadership

Research Funding & Awards: Mitacs Accelerate Industrial Fellowship with Ciena (\$10K) • Graduate Research Funding (\$29,348) • Match of Minds Research Scholarship (sole departmental recipient) • Brock Entrance Scholarship (\$6K)

Teaching: Teaching Assistant — COSC 3P97 Mobile Computing & COSC 3P95 Software Analysis & Testing (2025–26)